Lab 02: Zener Diode – Load Line Regulation

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Coded in LaTeX

Contents

1 Abstract

This experiment investigates the voltage regulation properties of a Zener diode in both unloaded and loaded conditions. A 12 V Zener diode was biased through a series resistor and tested under three configurations: no external load, a moderate 3 kΩ load, and a heavy 470 Ω load. Measured data showed that the Zener maintained approximately 11.5 V under no-load conditions, with current through the diode increasing slightly as the input voltage rose. When resistive loads were applied, however, the diode dropped out of regulation, supplying negligible current and allowing the output voltage to fall below the nominal breakdown voltage. Simulated results matched closely with experimental observations, validating the diode model and highlighting the limitations of Zener regulation when the series resistor cannot support both the load current and the Zener bias. The experiment demonstrates the importance of proper resistor sizing in maintaining Zener operation within its breakdown region.

2 Introduction

The Zener diode is widely used as a voltage reference and regulator because of its nearly constant reverse breakdown voltage over a broad range of current. This lab focuses on exploring the Zener's regulation behavior under different loading conditions and comparing experimental results with circuit simulations.

3 Objective

The objective of this lab is to investigate Zener diode voltage regulation, demonstrate the impact of load resistance on regulation performance, and compare measured results to simulated results using a custom diode model.

4 Theory

The almost vertical drop of the i-v curve of the Zener diode in the breakdown region suggests it can be used as a voltage regulator. A voltage regulator intends to provide a constant DC output voltage, irrespective of:

- **Load Regulation:** Variation in load current
- **Line Regulation:** Variation in supply voltage

For current smaller than the knee current ($-I_{ZK}$), the current remains almost constant. V_Z and I_{ZT} are test parameters provided by the manufacturer. As the current deviates from the test current I_{ZT}, the voltage also changes, resulting in a small resistance, $r_Z = \Delta V/\Delta I$. r_Z is typically in the range of a few ohms to tens of ohms; the lower the value of r_Z, the more stable the performance of the Zener diode.

$$\Delta V = \Delta I \cdot r_Z$$

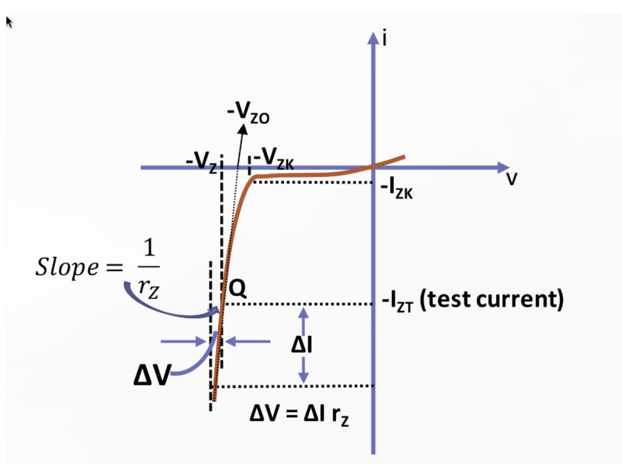

Figure 1: Zener diode i-v characteristics

The almost linear i-v characteristic of the Zener diode allows it to be modeled as shown in Figure 2, with a constant voltage source (V_{ZO}) in series with r_Z. From this representation:

$$V_Z = V_{ZO} + r_Z I_Z, \quad \text{for } I_Z > I_{ZK}, V_Z > V_{ZO}$$

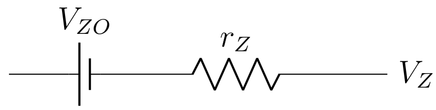

Figure 2: Small-signal Zener diode equivalent circuit

5 Experimental Procedures

The following circuits were utilized in order to determine the characteristics of the Zener diode in an introductory fashion.

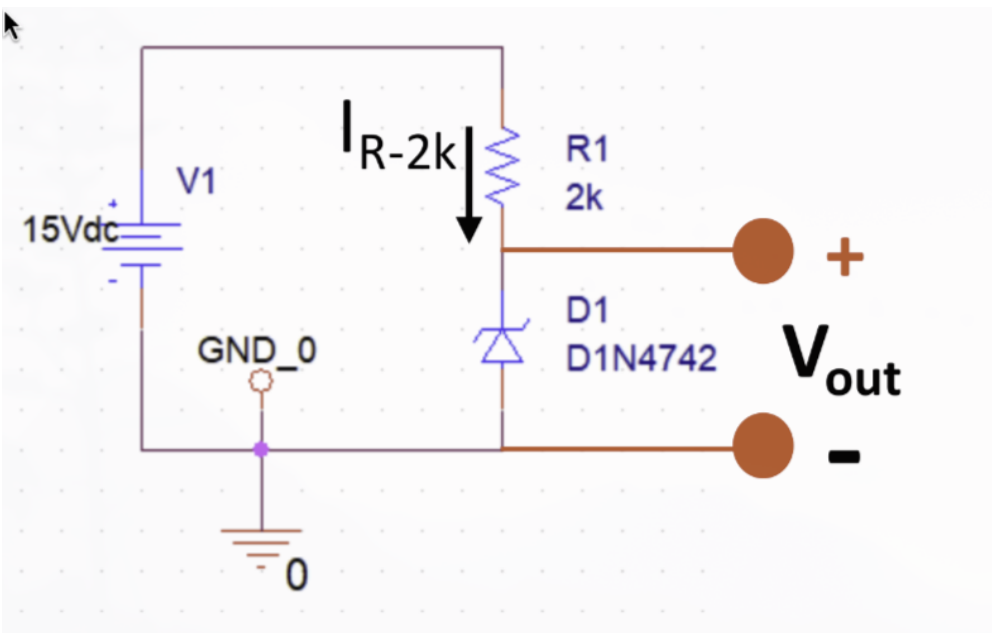

Figure 3: Circuit 1 – No Load

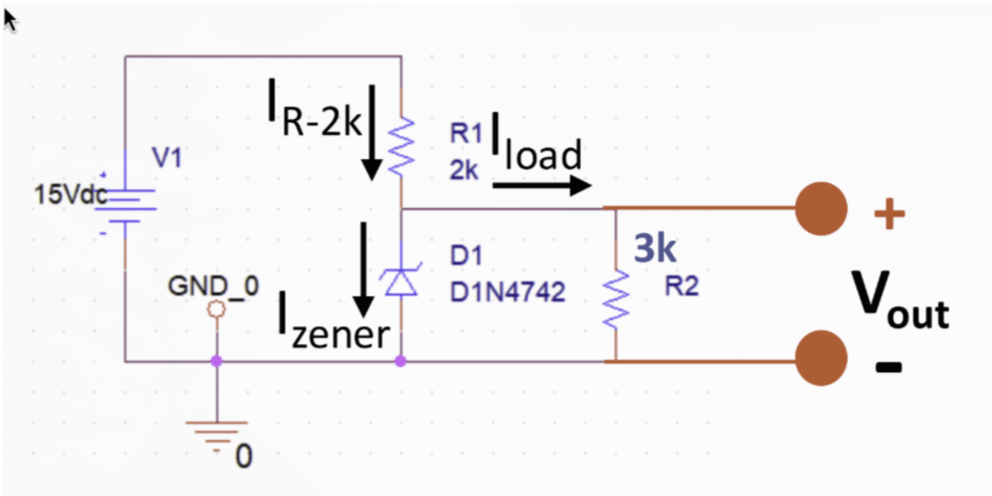

Figure 4: Circuit 2/3 – With Resistive Loads

Table 1: Reistor Values

Resistor	Measured Value
$2K\Omega$	$2.00K\Omega$
$3K\Omega$	$3.01K\Omega$
$0.47K\Omega$	$0.470K\Omega$

6 Results

6.1 Circuit 1 - No Load Condition

Table 2: Circuit 1 - No Load

V_{IN}	**14V**	**15V**	**16V**
$V_{2K\Omega}$	2.54V	3.53V	4.52V
V_{Zener}	11.48V	11.49V	11.50V
I_{Zener}	1.28mA	1.77mA	2.27mA

6.2 Circuit 2 - $3K\Omega$ Load

Table 3: Circuit 2 - 3K Load

V_{IN}	**14V**	**15V**	**16V**
$V_{2K\Omega}$	5.58V	5.98V	6.38V
V_{Zener}	8.43V	9.03V	9.63V
$I_{R2K\Omega}$	2.80mA	3.00mA	3.20mA
I_{load}	2.80mA	3.00mA	3.20mA
I_{Zener}	0.2 μA	0.2 μA	0.2 μA

6.3 Circuit 3 - $0.47K\Omega$ Load

Table 4: Circuit 3 - 0.47K Load

V_{IN}	**14V**	**15V**	**16V**
$V_{2K\Omega}$	11.33V	12.07V	12.94V
V_{Zener}	2.68V	2.87V	3.07V
$I_{R2K\Omega}$	5.70mA	6.10mA	6.52mA
I_{load}	5.70mA	6.10mA	6.52mA
I_{Zener}	0.2 μA	0.2 μA	0.2 μA

6.4 Line Regulation Analysis

The variation in Zener voltage with respect to supply voltage was measured experimentally for three different load conditions:

- **No Load:** V_{IN}: 14→16 V, V_Z: 11.48→11.50 V.
  Slope $\Delta V_Z/\Delta V_{IN} = 0.01$ V/V (10 mV per volt).
  Percentage change over the span: $(0.02/11.48) \times 100\% \approx 0.17\%$.

- **3 kΩ Load:** V_Z: 8.43→9.63 V for the same input change.
  Slope = 0.60 V/V (poor regulation; Zener off).

- **470 Ω Load:** V_Z: 2.68→3.07 V.
  Slope = 0.20 V/V (dominated by divider action; Zener off).

6.5 Load Regulation Analysis

As load current increased (no load → 3 kΩ → 470 Ω), the Zener voltage fell from ∼11.5 V to ∼9 V and then to ∼3 V. This shows that the series resistor must provide enough current to satisfy both the load and a minimum Zener current (above I_{ZK}) or regulation is lost.

6.6 Derived Device Parameters

From the no-load measurements around 14–16 V:

$$\Delta V_Z \approx 0.02 \text{ V}, \quad \Delta I_Z \approx 2.27 - 1.28 = 0.99 \text{ mA}$$

$$\Rightarrow \; r_Z \approx \frac{\Delta V_Z}{\Delta I_Z} \approx \frac{0.02}{0.00099} \approx 20 \ \Omega$$

This is larger than the datasheet $r_{ZT} \approx 9\ \Omega$ at $I_{ZT} = 21$ mA, as expected, since our operating current (1–2 mA) is well below the test current and the Zener slope is steeper (higher r_Z) at lower currents.

Voltage Variation vs. V_{IN} (measured)

- No load: $\Delta V_Z/\Delta V_{IN} = 0.01$ V/V (10 mV/V).
- 3 kΩ load: $\Delta V_Z/\Delta V_{IN} = 0.60$ V/V.
- 470 Ω load: $\Delta V_Z/\Delta V_{IN} = 0.20$ V/V.

Zener Current Variation vs. V_{IN} (measured)

- No load: I_Z: 1.28→2.27 mA over $\Delta V_{IN} = 2$ V $\Rightarrow \Delta I_Z/\Delta V_{IN} \approx 0.99/2 = 0.495\,\text{mA/V}$.
- 3 kΩ load: $I_Z \approx 0.2\,\mu\text{A}$ (constant) $\Rightarrow$ slope ≈ 0.
- 470 Ω load: $I_Z \approx 0.2\,\mu\text{A}$ (constant) $\Rightarrow$ slope ≈ 0.

6.7 Simulated Portion

The following netlist was used to produce the simulated data. For the 3 kΩ case, set `R2 N002 0 3k`; for the no-load case, omit `R2`.

```
* Lab 02 - Zener Diode -- Load Line Regulation
* Arturo Salinas-Aguayo
* 05 September 2025
* @Begin
V1 N001 0 14
D1 0 N002 D1N4742
R1 N001 N002 2k
R2 N002 0 0.47k
.dc v1 14 16 .5
.model D1N4742 D( BV=12  IBV=0.021  NBV=7.31  RS=0.5
+ IS=1e-9  N=1.5  CJO=80p  VJ=0.7  M=0.33  TT=50n )
.end
* @End
```

- Instantiates a DC source, Zener diode (custom model), and resistors.

- Performs a DC sweep in 0.5 V steps from 14 V to 16 V.
- Note: Negative simulated I_{Zener} values arise from the standard device current reference direction (from anode to cathode).

6.8 Simulation Data - Circuit 1 – No Load

Table 5: Simulated Circuit 1 - No Load

V_{IN}	**14V**	**15V**	**16V**
V_{Zener}	11.469V	11.529V	11.576V
I_{Zener}	-1.265mA	-1.735mA	-2.212mA

6.9 Simulation Data - Circuit 2 – $3K\Omega$ Load

Table 6: Simulated Circuit 2 - $3K\Omega$ Load

V_{IN}	**14V**	**15V**	**16V**
V_{Zener}	8.399V	8.999V	9.599V
I_{Zener}	-1.121nA	-3.708nA	-65.472nA

6.10 Simulation Data - Circuit 3 – $0.47K\Omega$ Load

Table 7: Simulated Circuit 3 - $0.47K\Omega$ Load

V_{IN}	**14V**	**15V**	**16V**
V_{Zener}	2.664V	2.854V	3.044V
I_{Zener}	-1.00nA	-1.00nA	-1.00nA

6.11 Summary

The experimental data confirm the theoretical behavior: under no load, the Zener provides excellent line regulation, but with significant loading the diode drops out of conduction. Proper resistor sizing and supply design are therefore critical to maintain regulation.

7 Conclusion

The results confirm that Zener diodes regulate effectively only when biased with sufficient current. In the no-load condition, the Zener voltage remained stable near 11.5 V, with the diode absorbing excess current from the series resistor. Under a 3 kΩ load, however, nearly all current was diverted to the load, causing the Zener to conduct almost no current and the regulated voltage to drop to the 8–9 V range. With a heavier 470 Ω load, the diode was fully out of regulation, and the voltage collapsed to 2–3 V. Simulation data reproduced these behaviors, reinforcing the experimental observations. The key design insight is that the series resistor and supply voltage must be chosen such that the diode maintains a bias current above its minimum value even under maximum load. Without this condition, the Zener cannot act as a reliable voltage regulator.